

Program 6: Deploy an Automated Build Pipeline using Docker Hub

Project Structure

DEVOPSLAB

```
--prg6
|--Dockerfile
|--package.json
|--package-lock.json
|--node_modules
```

Step 1: Create a GitHub Personal Access Token

Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk ().*

1 **General**

Owner * /
program6 is available.

Great repository names are short and memorable. How about [legendary-tribble](#)?

Description
0 / 350 characters

2 **Configuration**

Choose visibility * Choose who can see and commit to this repository

Add README ☐ READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore .gitignore tells git which files not to track. [About ignoring files](#)

Add license Licenses explain how others can use your code. [About licenses](#)

Step 2: Create a Dockerfile and build the image

```
1rv24mc054_khushijs@khushi-GF63-Thin-105CXR: ~/DEVOPSLAB/prg6
GNU nano 7.2 Dockerfile
FROM node:18

WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["npm", "start"]
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ npm init -y
Wrote to /home/1rv24mc054_khushijs/DEVOPSLAB/prg6/package.json:
```

```
{
  "name": "prg6",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ npm install express
```

```
added 65 packages, and audited 66 packages in 5s
```

```
22 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ npm install
```

```
up to date, audited 66 packages in 1s
```

```
22 packages are looking for funding
  run `npm fund` for details
```

```
found 0 vulnerabilities
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ sudo docker build -t program6 .
```

```
[+] Building 33.7s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 144B
=> [internal] load metadata for docker.io/library/node:18
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
=> => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
=> [internal] load build context
=> => transferring context: 2.16MB
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:a766f5e6ba32485051bc89b59f6563ed7e74a2e9238eda6c4aeb6ec0c5c5ca9f
=> => exporting config sha256:be3ac5d642f045127ad5728164889f7c551aa0e36ea9cd41bd34a25c6b45c5d6
=> => exporting attestation manifest sha256:4aa32a60ab39ad39c324bd66641d6050b9902f24d83e10853bbb0bf5e751c1c0
=> => exporting manifest list sha256:9183ba0849b0face81aa17e5dc859d6e733e96f725fad687f03873827e8fcc25
=> => naming to docker.io/library/program6:latest
=> => unpacking to docker.io/library/program6:latest
docker:default
0.1s
0.0s
2.0s
0.1s
0.0s
0.4s
0.3s
0.3s
0.1s
0.0s
1.2s
6.5s
13.3s
8.5s
4.4s
0.2s
0.3s
1.2s
0.3s
0.0s
1.7s
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$
```

Step 3: Tag the Image for Docker Hub

Step 4: Push the Image to Docker Hub

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ docker tag program6 khushijs/program6:latest
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ docker login -u khushijs
```

 Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:

WARNING! Your credentials are stored unencrypted in '/home/1rv24mc054_khushijs/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ docker push khushijs/program6:latest
The push refers to repository [docker.io/khushijs/program6]
c6b30c3f1696: Pushed
32723d1ab310: Layer already exists
37927ed901b1: Pushed
462fec842d09: Pushed
3697be50c98b: Layer already exists
e23f099911d6: Pushed
3e6b9d1a9511: Pushed
461077a72fb7: Layer already exists
17956521f473: Pushed
97a7198558ad: Pushed
8c9e5eaf2530: Pushed
79b2f47ad444: Pushed
cda7f44f2bdd: Layer already exists
latest: digest: sha256:9183ba0849b0face81aa17e5dc859d6e733e96f725fad687f03873827e8fcc25 size: 856
```

Step 5: Tag the Image for GitHub Container Registry(GHCR)

Step 6: Push the Image to GHCR

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ docker tag program6 ghcr.io/khushijsampagavi/program6:latest
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ echo 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 | docker login ghcr.io
-u /khushijsampagavi --password-stdin
```

WARNING! Your credentials are stored unencrypted in '/home/1rv24mc054_khushijs/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

```
Login Succeeded
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ docker push ghcr.io/khushijsampagavi/program6:latest
The push refers to repository [ghcr.io/khushijsampagavi/program6]
462fec842d09: Pushed
32723d1ab310: Pushed
17956521f473: Pushed
3697be50c98b: Pushed
37927ed901b1: Pushed
c6b30c3f1696: Pushed
97a7198558ad: Pushed
461077a72fb7: Pushed
e23f099911d6: Pushed
79b2f47ad444: Pushed
cda7f44f2bdd: Pushed
8c9e5eaf2530: Pushed
3e6b9d1a9511: Pushed
latest: digest: sha256:9183ba0849b0face81aa17e5dc859d6e733e96f725fad687f03873827e8fcc25 size: 856
```

Step 7: Connect the Local Project to GitHub

Step 8: Push the Source Code to GitHub

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ git remote add origin https://github.com/khushijsampagavi/program6.git
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ git add .
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ git commit -m "Initial commit"
[master (root-commit) 59776a2] Initial commit
588 files changed, 61767 insertions(+)
create mode 100644 Dockerfile
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json
create mode 100644 node_modules/call-bind-apply-helpers/.eslintrc
```

```
1rv24mc054_khushijs@khushi-GF63-Thin-10SCXR:~/DEVOPSLAB/prg6$ git push -u origin main
Username for 'https://github.com': khushijsampagavi
Password for 'https://khushijsampagavi@github.com':
Enumerating objects: 677, done.
Counting objects: 100% (677/677), done.
Delta compression using up to 8 threads
Compressing objects: 100% (626/626), done.
Writing objects: 100% (677/677), 668.80 KiB | 2.52 MiB/s, done.
Total 677 (delta 121), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (121/121), done.
To https://github.com/khushijsampagavi/program6.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

The screenshot shows the GitHub web interface for a repository named 'program6' owned by 'khushijsampagavi'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main'. The repository contains an initial commit with files: node_modules, Dockerfile, package-lock.json, and package.json. A 'README' section is visible with a prompt to 'Add a README'. The right sidebar shows sections for 'About', 'Releases', 'Packages', 'Languages' (Dockerfile 100.0%), and 'Suggested workflows' (Docker image).

The screenshot shows the Docker Hub 'My Hub' page for the user 'khushijs'. The page displays a list of repositories within the 'khushijs' namespace. A table lists the repository 'khushijs/program6', which was last pushed 25 minutes ago, contains an 'IMAGE', is public, and is currently 'Inactive'. The left sidebar shows navigation options like 'Repositories', 'Hardened Images', 'Collaborations', 'Settings', 'Default privacy', 'Notifications', 'Billing', 'Usage', 'Pulls', and 'Storage'.